Benchmark Comparison: AFFORD-Bench vs. LIBERO-PRO & LIBERO-X

Key Differences in Physical Affordances, Perceptual Modalities, Policy Universality, and Scale

RSS 2027 BENCHMARK

BENCHMARK DIMENSION Core Evaluation Axes	LIBERO-PRO Zhou et al., 2026 • Robustness	LIBERO-X Wang et al., 2026 • Language-X	★ AFFORD-Bench (Ours) Physics-Grounded Universal Benchmark
Contact Physics Friction & mass dynamics	Nominal (Fixed μ = 0.5)	Nominal (Fixed μ = 0.5)	✓ True Contact Pads (μ ∈ [0.12, 0.25] & Mass)
Perception Inputs Visual & geometric sensors	1× RGB (128 × 128)	2× RGB (256 × 256)	✓ Dual RGB (640×660 / 300×420) + 3D Depth
Policy Universality Policy & dataset formats	VLA Models Only	VLA Models Only	✓ Universal + LeRobot v2.0 & RLDS
Domain Scope Environment breadth & tasks	4 Suites (130 Tasks)	5 Suites (50 Tasks)	✓ 7 Diverse Domains (210 Tasks)
Precision & Anti-Cheat Tolerance & impulse probes	Coarse / Kinematic Welds	Coarse / Kinematic Welds	✓ 2–2.5 mm Tol. + Reactive Impulse Probe
Verified Solvability Empirical sanity baselines	Unverified Null Baselines	Unverified Null Baselines	✓ 99.8% Nominal Oracle (0.0% Null/Rand)
Benchmark Scale Normalized data volume	130 Tasks • 62% Scale	50 Tasks • 24% Scale	★ 210 Tasks • 100% (4.2× Scale)